

EXPERIMENT – 8**Server.js:**

```
const express = require("express");
const app = express();
const PORT = 3000;

app.use(express.json());

let students = [
  { id: 1, name: "Dhanush", rollNo: "160124748136", gpa: 8.21, semester: 5, mobileNo: "9876543210" },
  { id: 2, name: "Hemanth", rollNo: "160124748142", gpa: 8.76, semester: 5, mobileNo: "9123456780" },
  { id: 3, name: "Afridi", rollNo: "160124748157", gpa: 7.95, semester: 3, mobileNo: "9988776655" },
  { id: 4, name: "Tanish", rollNo: "160124748163", gpa: 8.48, semester: 3, mobileNo: "9012345678" },
  { id: 5, name: "Hanmanthu", rollNo: "160124748174", gpa: 8.92, semester: 4, mobileNo: "9345678901" },
  { id: 6, name: "Karthik", rollNo: "160124748189", gpa: 8.35, semester: 4, mobileNo: "9867452310" }
];

// Home
app.get("/", (req, res) => {
  res.send("Student Management REST API is running");
});

// GET - All students
app.get("/api/students", (req, res) => {
  res.json(students);
});

// GET - Student by Roll No
app.get("/api/students/:rollNo", (req, res) => {
  const student = students.find(s => s.rollNo === req.params.rollNo);

  if (!student)
    return res.status(404).json({ message: "Student not found" });

  res.json(student);
});

// POST - Add student
app.post("/api/students", (req, res) => {
  const { name, rollNo, gpa, semester, mobileNo } = req.body;

  if (!name || !rollNo || !gpa || !semester || !mobileNo)
    return res.status(400).json({ message: "All details are required" });

  const student = {
    id: students.length + 1,
    name, rollNo, gpa, semester, mobileNo
  };
  students.push(student);
  res.status(201).json(student);
});
```

```

    });

    students.push(student);

    res.status(201).json({
      message: "Student added successfully",
      student
    });
  });

// PUT - Update student
app.put("/api/students/:rollNo", (req, res) => {
  const student = students.find(s => s.rollNo === req.params.rollNo);

  if (!student)
    return res.status(404).json({ message: "Student not found" });

  const { name, gpa, semester, mobileNo } = req.body;

  student.name = name || student.name;
  student.gpa = gpa || student.gpa;
  student.semester = semester || student.semester;
  student.mobileNo = mobileNo || student.mobileNo;

  res.json({
    message: "Student updated successfully",
    student
  });
});

// PUT - Promote student
app.put("/api/students/:rollNo/promote", (req, res) => {
  const student = students.find(s => s.rollNo === req.params.rollNo);

  if (!student)
    return res.status(404).json({ message: "Student not found" });

  if (student.semester >= 8)
    return res.status(400).json({ message: "Already in final semester" });

  student.semester++;

  res.json({
    message: "Student promoted successfully",
    student
  });
});

// DELETE - Delete student
app.delete("/api/students/:rollNo", (req, res) => {
  const index = students.findIndex(s => s.rollNo === req.params.rollNo);

  if (index === -1)
    return res.status(404).json({ message: "Student not found" });

  const student = students.splice(index, 1)[0];

  res.json({
    message: "Student deleted successfully",

```

Laboratory Record

of FSD LAB

Roll No.: 160124748101

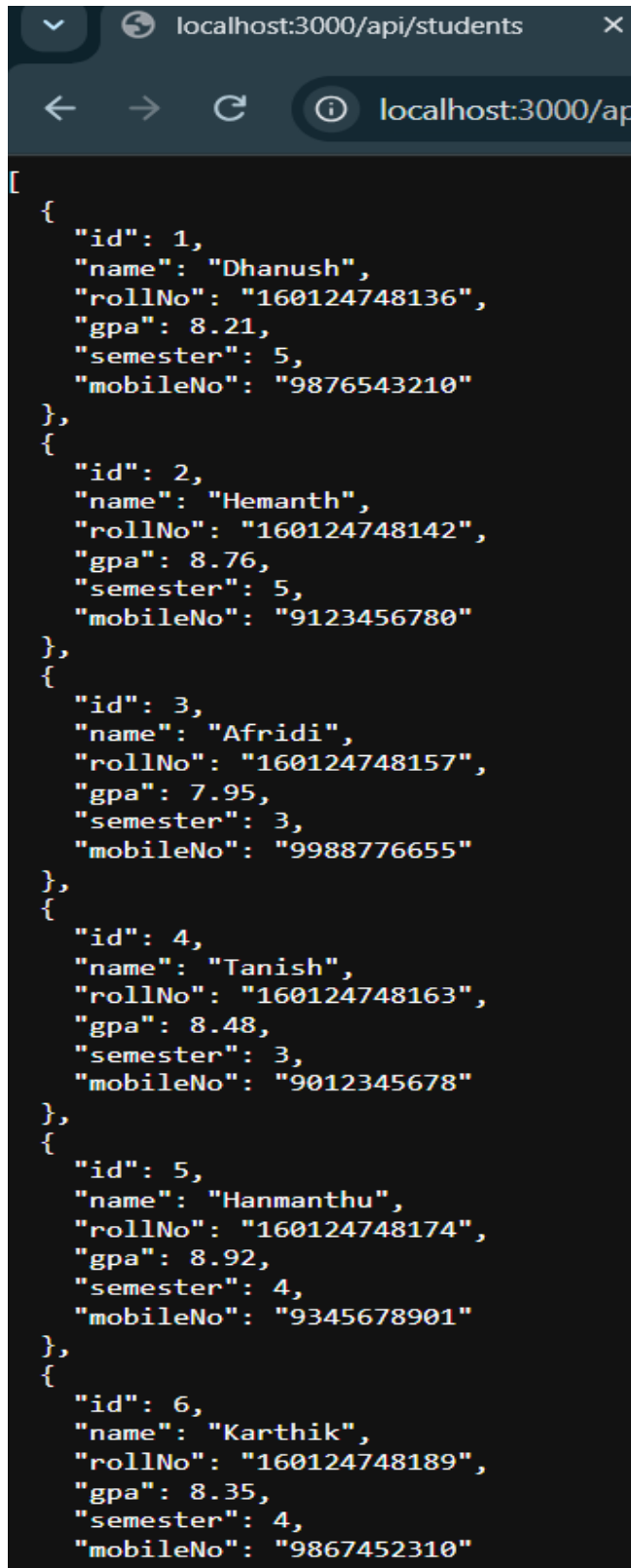
Experiment No.: 08

Sheet No.: _____

Date: 26/09/26

```
        student
    });
});

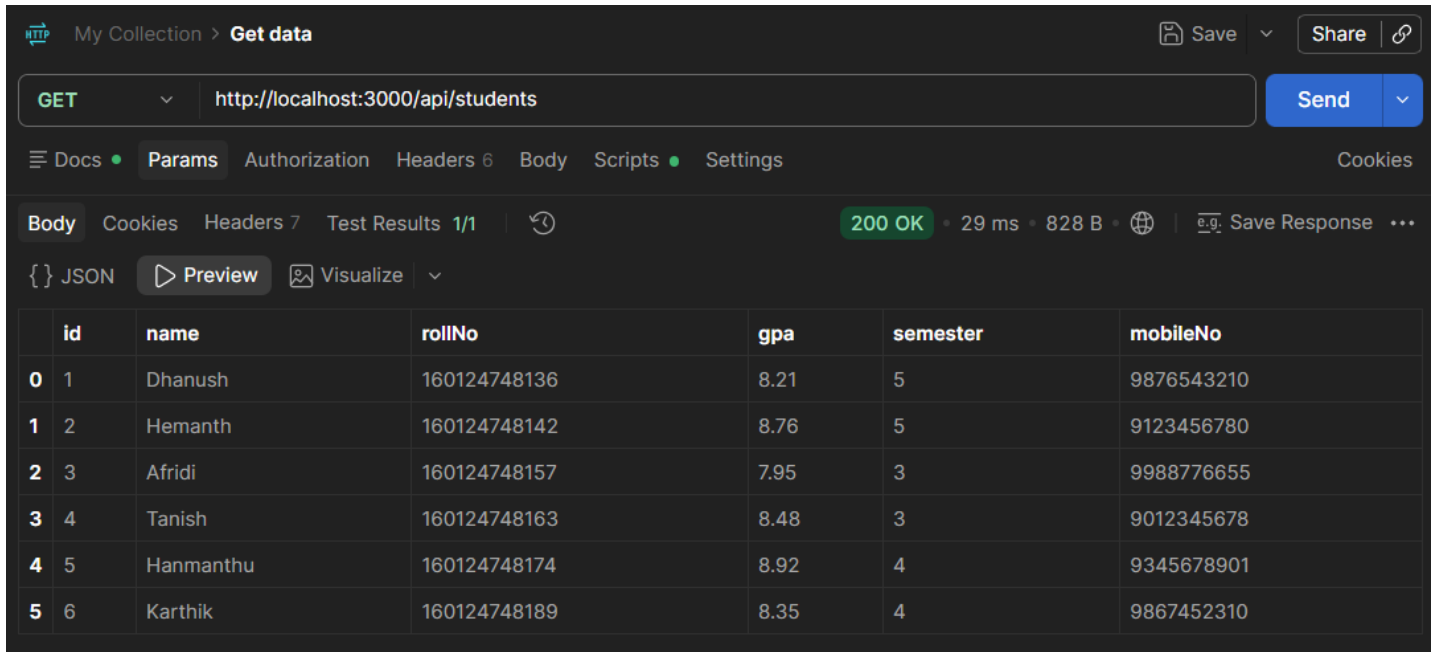
app.listen(PORT, () => {
    console.log(`Server running at http://localhost:${PORT}`);
});
```



```
[
  {
    "id": 1,
    "name": "Dhanush",
    "rollNo": "160124748136",
    "gpa": 8.21,
    "semester": 5,
    "mobileNo": "9876543210"
  },
  {
    "id": 2,
    "name": "Hemanth",
    "rollNo": "160124748142",
    "gpa": 8.76,
    "semester": 5,
    "mobileNo": "9123456780"
  },
  {
    "id": 3,
    "name": "Afridi",
    "rollNo": "160124748157",
    "gpa": 7.95,
    "semester": 3,
    "mobileNo": "9988776655"
  },
  {
    "id": 4,
    "name": "Tanish",
    "rollNo": "160124748163",
    "gpa": 8.48,
    "semester": 3,
    "mobileNo": "9012345678"
  },
  {
    "id": 5,
    "name": "Hanmanthu",
    "rollNo": "160124748174",
    "gpa": 8.92,
    "semester": 4,
    "mobileNo": "9345678901"
  },
  {
    "id": 6,
    "name": "Karthik",
    "rollNo": "160124748189",
    "gpa": 8.35,
    "semester": 4,
    "mobileNo": "9867452310"
  }
]
```

1) GET OPERATION:

i) To GET all Students:



My Collection > Get data

GET `http://localhost:3000/api/students` Send

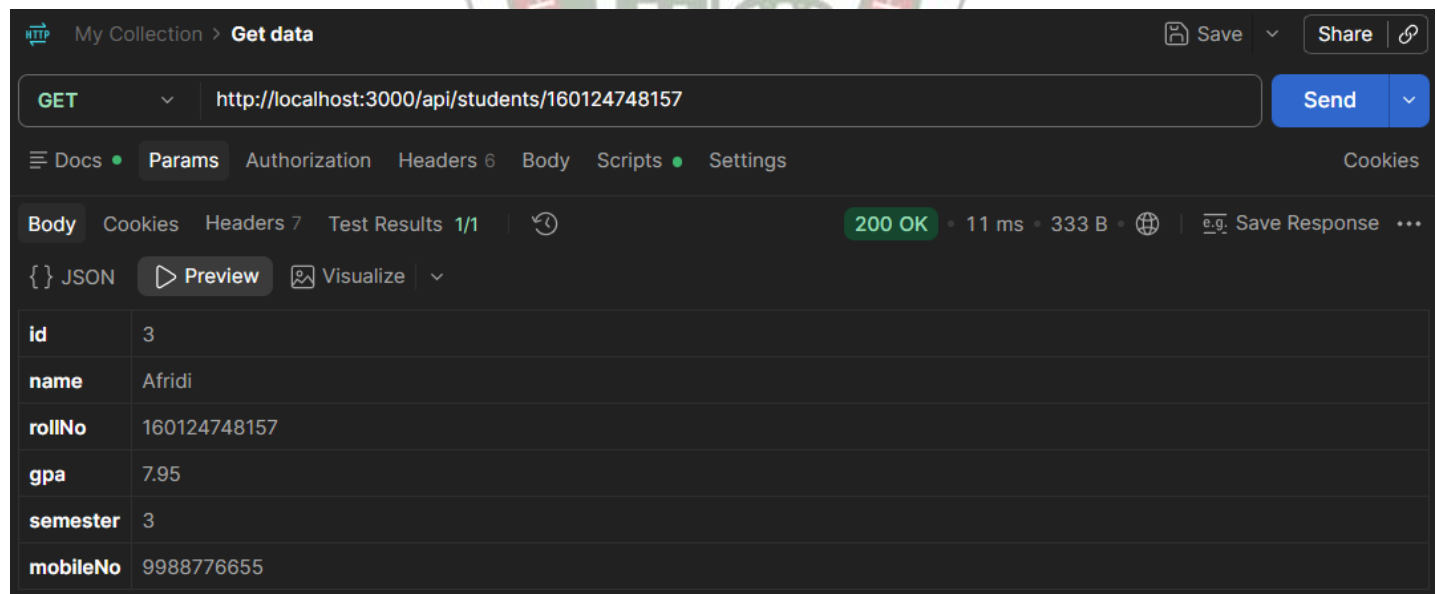
Docs Params Authorization Headers 6 Body Scripts Settings Cookies

Body Cookies Headers 7 Test Results 1/1 200 OK • 29 ms • 828 B • Save Response

{ } JSON Preview Visualize

	id	name	rollNo	gpa	semester	mobileNo
0	1	Dhanush	160124748136	8.21	5	9876543210
1	2	Hemanth	160124748142	8.76	5	9123456780
2	3	Afridi	160124748157	7.95	3	9988776655
3	4	Tanish	160124748163	8.48	3	9012345678
4	5	Hanmanthu	160124748174	8.92	4	9345678901
5	6	Karthik	160124748189	8.35	4	9867452310

ii) To GET Student by Roll.no:



My Collection > Get data

GET `http://localhost:3000/api/students/160124748157` Send

Docs Params Authorization Headers 6 Body Scripts Settings Cookies

Body Cookies Headers 7 Test Results 1/1 200 OK • 11 ms • 333 B • Save Response

{ } JSON Preview Visualize

id	3
name	Afridi
rollNo	160124748157
gpa	7.95
semester	3
mobileNo	9988776655

Laboratory Record

of FSD LAB

Roll No.: 160124748101

Experiment No.: 08

Sheet No.: _____

Date: 26/09/26

2) POST OPERATION:

POST http://localhost:3000/api/students **Send**

Docs Params Authorization Headers 8 **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** Schema Beautify

```
1 {
2   "name": "Rohit",
3   "rollNo": "160124748195",
4   "gpa": 8.15,
5   "semester": 3,
6   "mobileNo": "9876123450"
7 }
```

Body Cookies Headers 7 Test Results 0/1 201 Created • 927 ms • 389 B • Save Response

JSON Preview Visualize

```
1 {
2   "message": "Student added successfully",
3   "student": {
4     "id": 7,
5     "name": "Rohit",
6     "rollNo": "160124748195",
7     "gpa": 8.15,
8     "semester": 3,
9     "mobileNo": "9876123450"
10  }
11 }
```

3) PUT OPERATION:

PUT http://localhost:3000/api/students/160124748136 **Send**

Docs Params Authorization Headers 8 **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** Schema Beautify

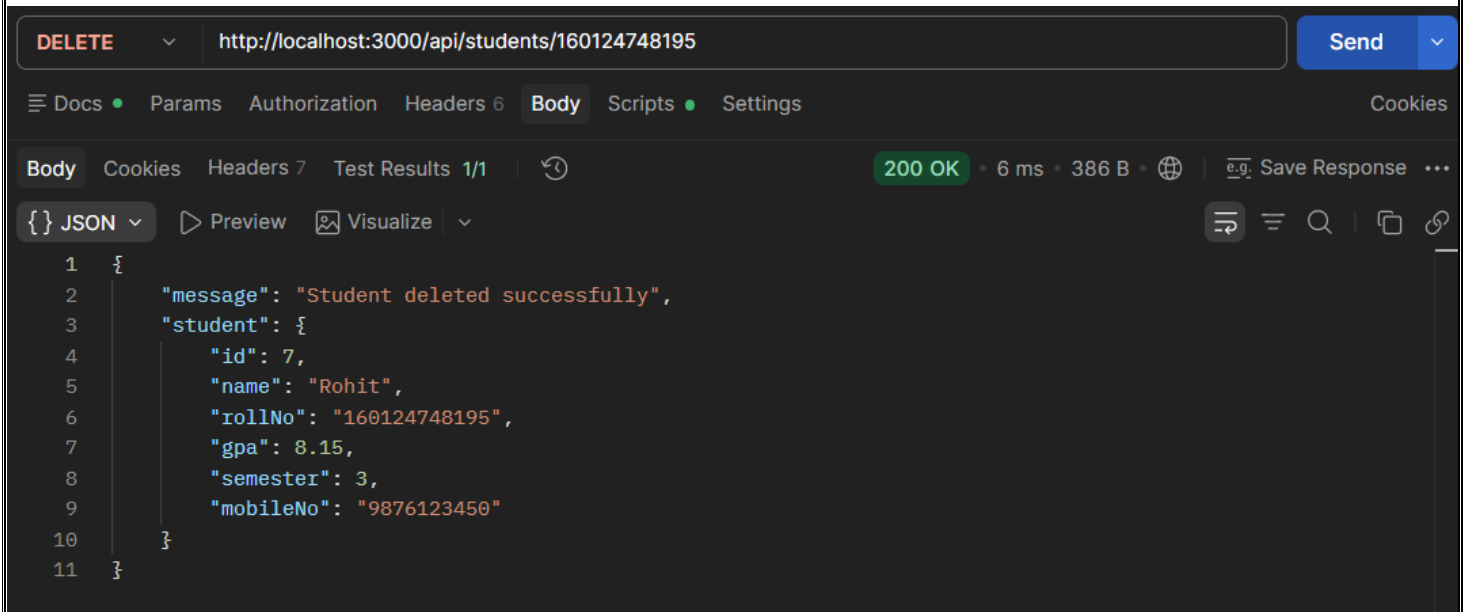
```
1 {
2   "name": "Dhanush",
3   "gpa": 8.75,
4   "semester": 5,
5   "mobileNo": "9876501234"
6 }
```

Body Cookies Headers 7 Test Results 1/1 200 OK • 6 ms • 388 B • Save Response

JSON Preview Visualize

```
1 {
2   "message": "Student updated successfully",
3   "student": {
4     "id": 1,
5     "name": "Dhanush",
6     "rollNo": "160124748136",
7     "gpa": 8.75,
8     "semester": 5,
9     "mobileNo": "9876501234"
10  }
11 }
```

4) DELETE OPERATION:



DELETE <http://localhost:3000/api/students/160124748195> Send

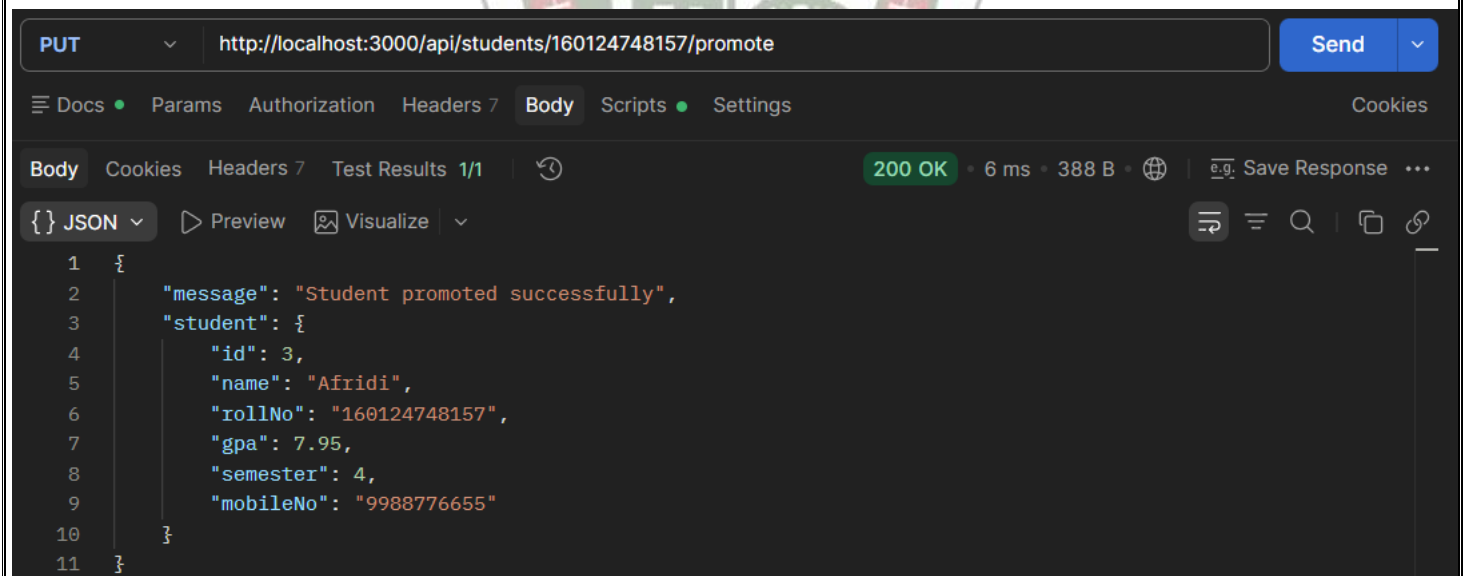
Docs Params Authorization Headers 6 Body Scripts Settings Cookies

Body Cookies Headers 7 Test Results 1/1 200 OK • 6 ms • 386 B • e.g. Save Response

{ } JSON Preview Visualize

```
1 {
2   "message": "Student deleted successfully",
3   "student": {
4     "id": 7,
5     "name": "Rohit",
6     "rollNo": "160124748195",
7     "gpa": 8.15,
8     "semester": 3,
9     "mobileNo": "9876123450"
10  }
11 }
```

5) STUDENT PROMOTION API:



PUT <http://localhost:3000/api/students/160124748157/promote> Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Body Cookies Headers 7 Test Results 1/1 200 OK • 6 ms • 388 B • e.g. Save Response

{ } JSON Preview Visualize

```
1 {
2   "message": "Student promoted successfully",
3   "student": {
4     "id": 3,
5     "name": "Afridi",
6     "rollNo": "160124748157",
7     "gpa": 7.95,
8     "semester": 4,
9     "mobileNo": "9988776655"
10  }
11 }
```